

RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

15th Sept 2021
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question on the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, **hand in your essay question SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 32
2	/ 8
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

This document consists of **15** printed pages.



Raffles Institution
Internal Examination

Section A

Answer **all** the questions in this section.

- 1** The COVID-19 pandemic is caused by the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), which was first identified in December 2019. As of June 2021, more than 180 million cases of infection and more than 3.9 million deaths have been confirmed worldwide.

The severity of COVID-19 symptoms is highly variable, ranging from asymptomatic to life-threatening. COVID-19 is transmitted via droplets in the air, small airborne particles and via contact with contaminated fluids.

Fig. 1.1 show the structure of the SARS-CoV-2 virus, which contains a single-stranded RNA genome.

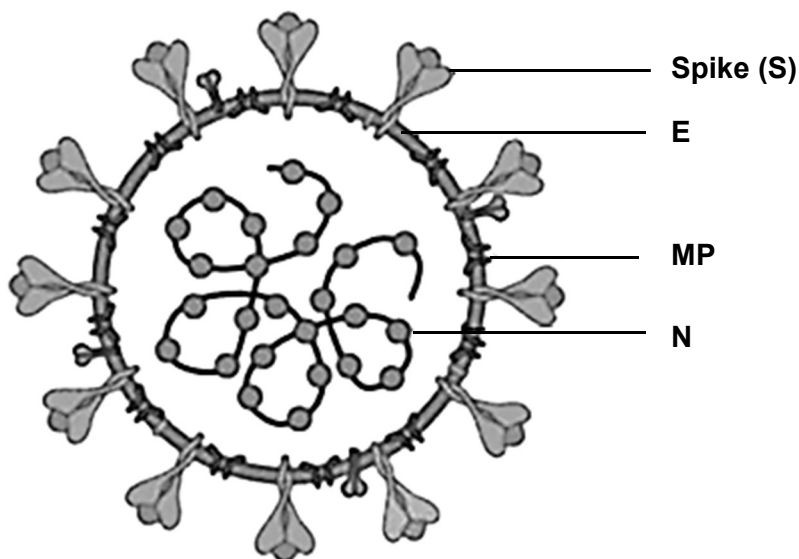


Fig.1.1

- (a)** Identify structure **E** and draw a labelled diagram showing its structure in the space below.

Structure **E**:

[3]

Fig. 1.2 shows the replication cycle of the SARS-CoV-2 virus.

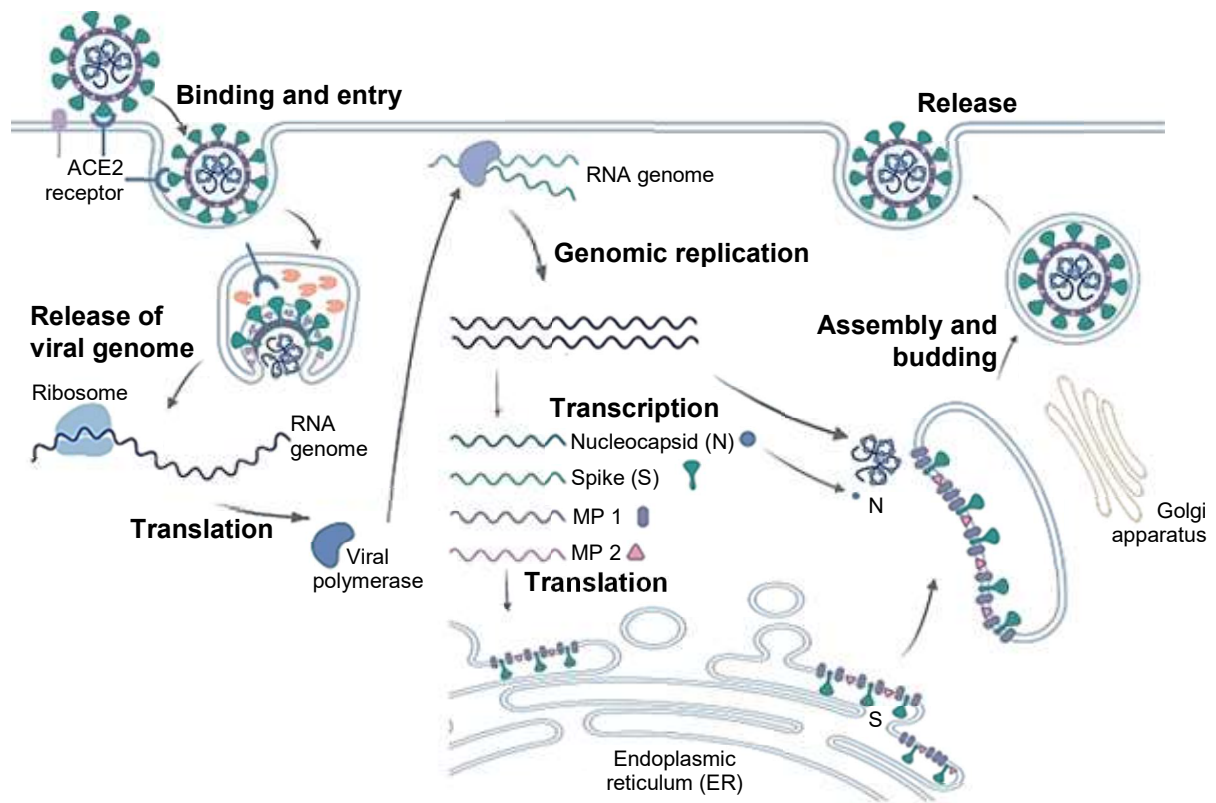


Fig. 1.2

- (b) Describe how genome replication differs between the SARS-CoV-2 virus and the T4 bacteriophage.

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The COVID-19 test is used to diagnose active infection with the SARS-CoV-2 virus.

The test involves the following steps:

- Insertion of a cotton swab through the patient's nostrils into the nasopharyngeal space so that a swab sample is collected.
- Extraction of viral genetic material using an extraction buffer.
- Reverse transcription and Polymerase Chain Reaction.

Fig. 1.3 below shows the process of (A) reverse transcription, followed by (B) modified PCR used in the COVID-19 test.

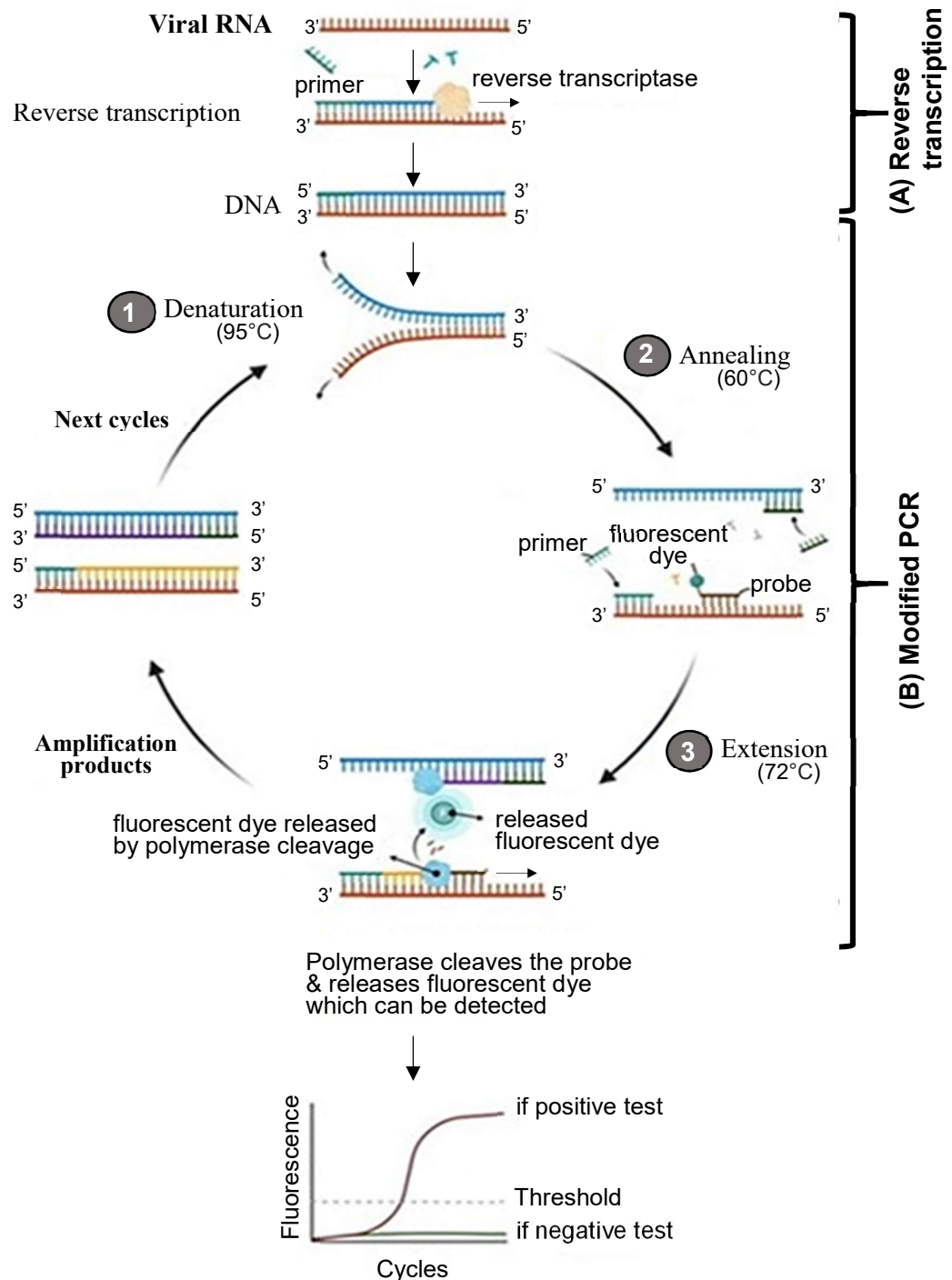


Fig. 1.3

- (c) (i) Explain why the (A) reverse transcription process is required before carrying out the (B) modified PCR.

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.....[2]

- (ii) With reference to the (B) modified PCR process shown in Fig. 1.3, explain the steps labelled:

denaturation;

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.....[2]

primer annealing.

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.....[2]

- (iii) Compare the primers and probes used in the (B) modified PCR process shown in Fig. 1.3.

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Scientists across the globe have worked rapidly to develop an effective vaccine against SARS-CoV-2. The SARS-CoV-2 spike protein has been identified as an effective target for use in vaccine design. Vaccines using the spike protein can be prepared via a variety of approaches.

Fig. 1.4 shows two different approaches.

- In the replication-defective viral vector vaccine, the DNA containing the gene that codes for the SARS-CoV-2 viral spike protein is enclosed in a viral vector capsid.
- In the lipid nanoparticle mRNA vaccine, the mature mRNA transcribed from the gene coding for SARS-CoV-2 viral spike protein is enclosed in a lipid nanoparticle coat.

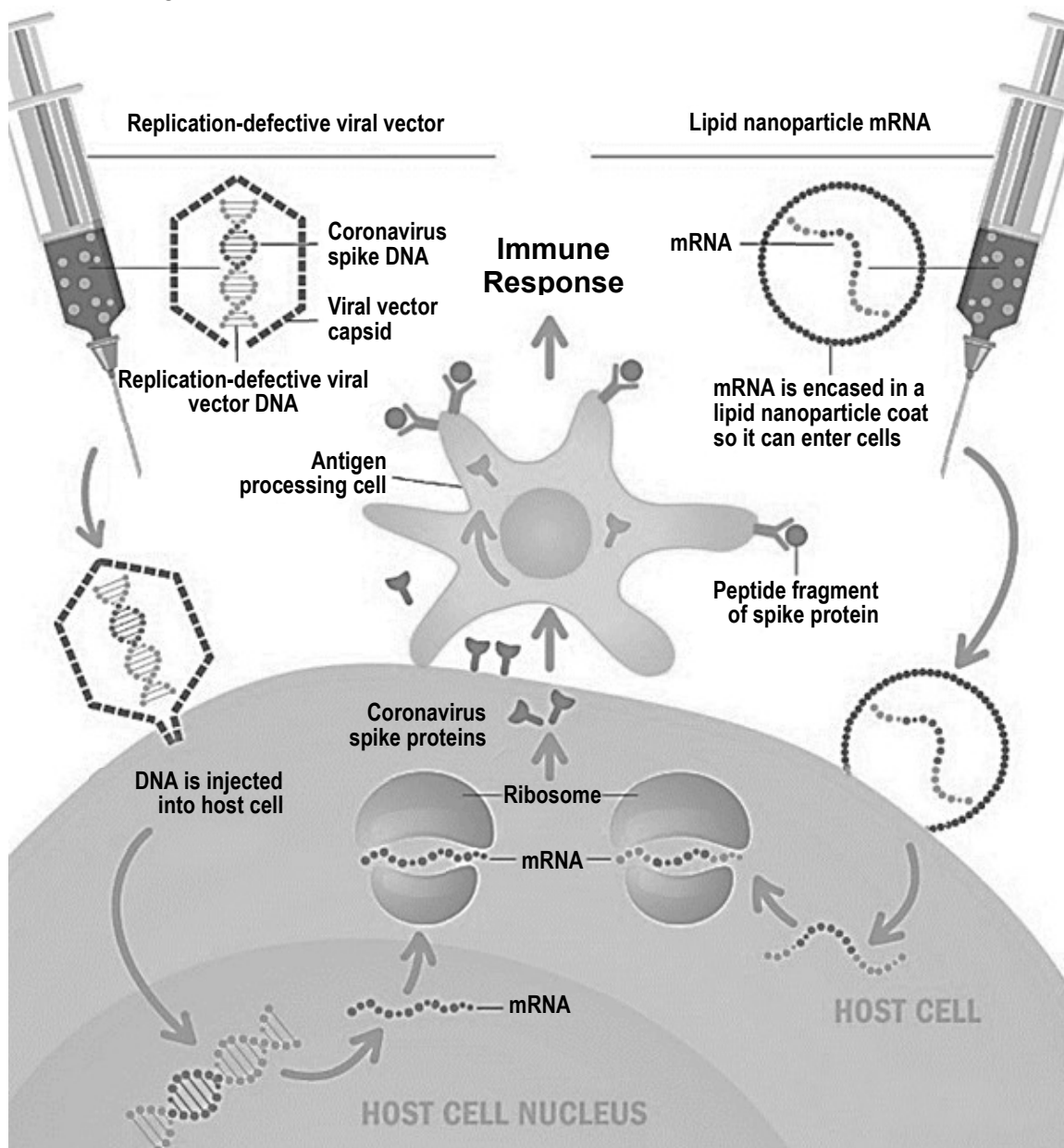


Fig. 1.4

- (d) (i) With reference to the lipid nanoparticle mRNA vaccine shown in Fig. 1.4, explain why mRNA is unable to enter the cell on its own but can do so when encased within a lipid nanoparticle coat.

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- (ii) Using your knowledge of mature mRNA, draw a labeled diagram to show the structure of the mature mRNA that was delivered into the cell.

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- (iii) Explain how the spike proteins are produced after the translation initiation complex is formed.

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- (iv) Suggest one advantage of using mRNA in the vaccine.

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- (v) Explain how the expression of spike proteins in a vaccinated individual protects against SARS-CoV-2 infection.

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- (e) From the time of the first discovery of the SARS-CoV-2, different variants have emerged. Among them, the Delta variant, also known as lineage B.1.617.2, is of international concern as it has been shown to have a higher transmission rate.

Sequence analysis revealed significant changes to the spike proteins. Table 1.1 shows these changes.

Table 1.1

position of amino acid	amino acid change	
	from	to
478	threonine	lysine
452	leucine	arginine
681	proline	arginine

- (i) Suggest and explain why these changes in amino acids have arisen.

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- (ii) Suggest an explanation for the increase in the Delta variant transmission rate.

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[Total: 32]

- 2 Fig. 2.1 and Fig. 2.2 show what happens when an artificially synthesized oligonucleotide binds to a specific sequence of DNA. This method silences the expression of the gene.

- (a) Fig. 2.1 shows an artificially synthesized oligonucleotide binding to the promoter region of the DNA sequence, forming a triple helix (triplex).

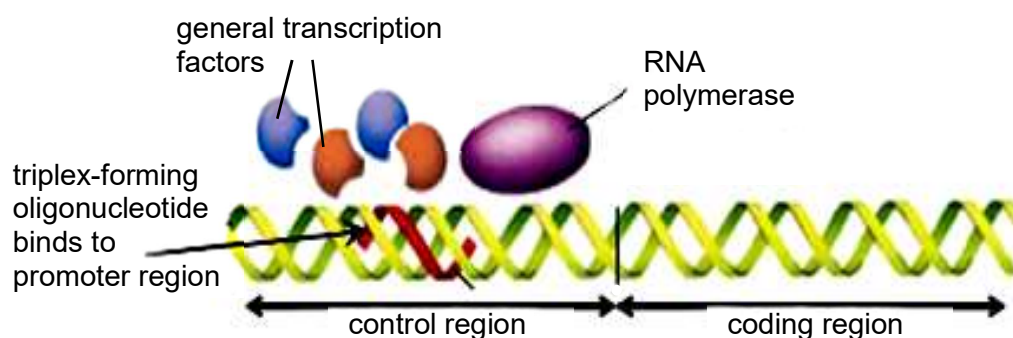


Fig. 2.1

- (i) With reference to Fig. 2.1, explain why transcription cannot occur in the presence of the artificially synthesised oligonucleotide.

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Fig. 2.2 shows another artificially synthesized oligonucleotide binding to the coding region of the DNA.

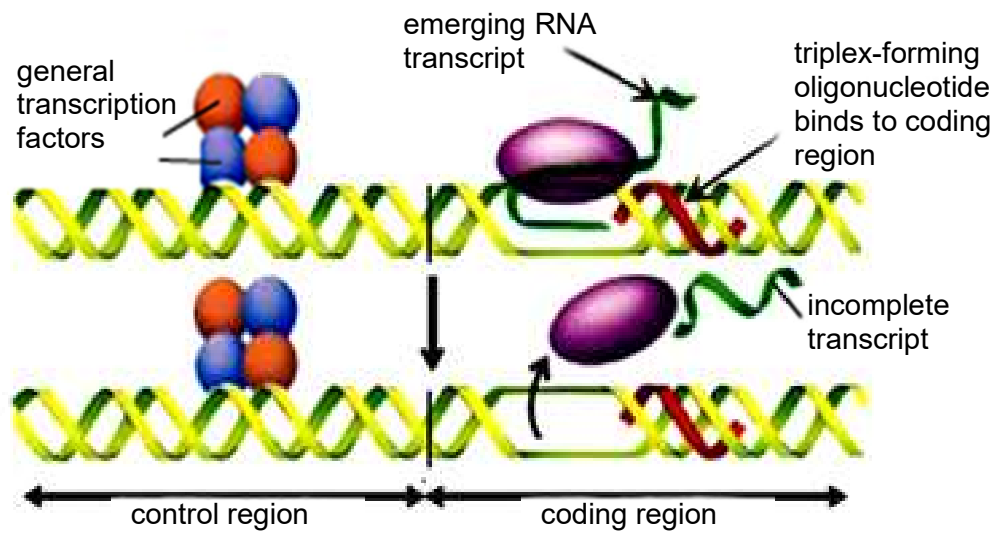


Fig. 2.2

(ii) Describe how gene silencing occurs in Fig. 2.2.

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Gene expression patterns can be analysed using polymerase chain reaction (PCR).

Methanolic extracts of *Imperata cylindrica* leaf (ICL) have been shown to affect the expression of the *CASP8* gene in human oral squamous carcinoma cells. *CASP8* gene codes for a caspase protein which plays a key role in activating programmed cell death. Human oral squamous carcinoma cells were treated with two different concentrations of methanolic extracts from ICL and the expression of the *CASP8* gene in these cells was measured.

The relative expression of the *CASP8* gene was measured from oral squamous carcinoma cells without treatment (control), with 320 µg/ml ICL treatment and with 640 µg/ml ICL treatment.

The relative expression of the *CASP8* gene can be seen in Fig. 2.3.

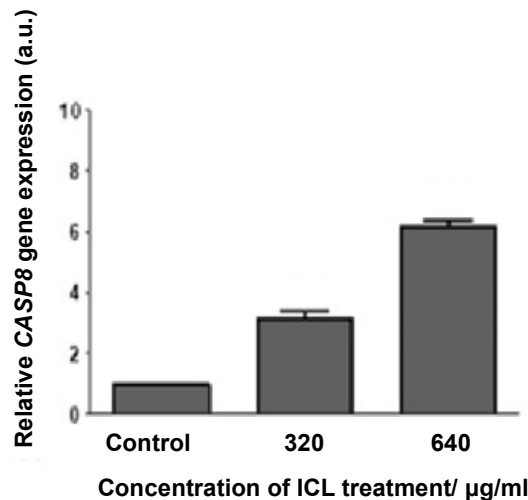


Fig. 2.3

- (b) (i) Using the information in Fig. 2.3, describe the results of ICL treatment on the human oral squamous carcinoma cells.

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[2]

- (ii) Explain the effectiveness of ICL in the treatment of human oral squamous carcinoma.

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[2]

[Total:8]

- 3 Mitochondrial DNA (mtDNA) is a tool for determining the origin of different populations. It is also an important object of study in different fields such as medical genetics, genetic genealogy and evolutionary phylogeny.

MELAS syndrome is an inherited disease in humans caused by a mutation in a gene located in the mtDNA.

Fig. 3.1 shows four generations of a family where several individuals are affected by MELAS syndrome.

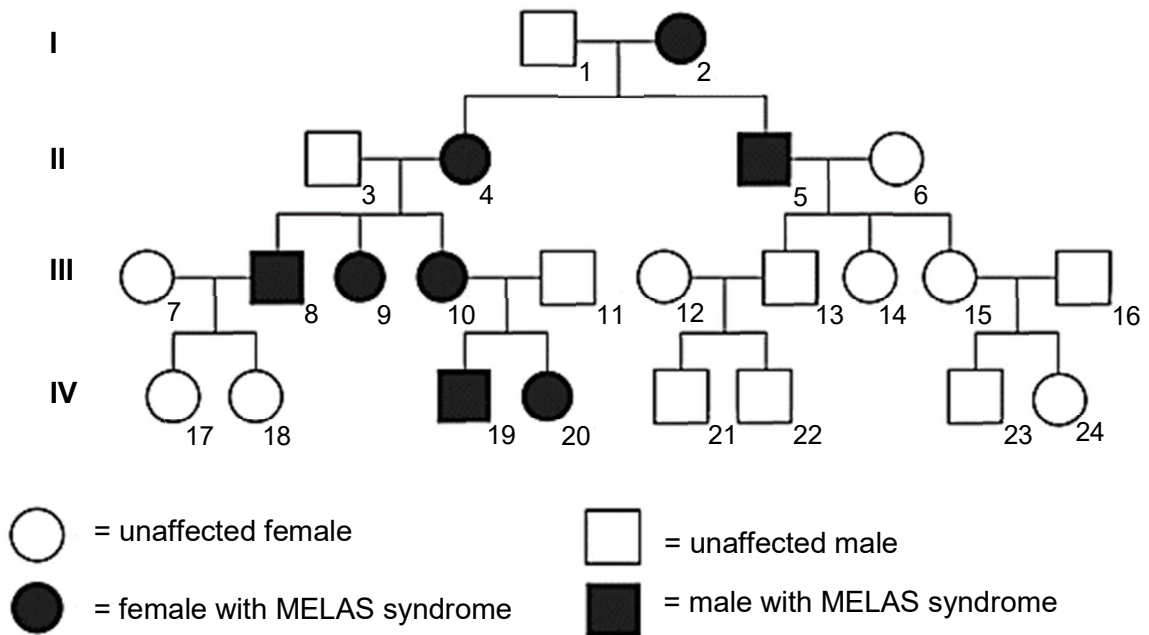


Fig. 3.1

- (a) (i) Explain, using evidence from Fig. 3.1, why MELAS syndrome is a mtDNA disease.

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- (ii) Describe and explain characteristics of mtDNA that makes it more useful than using nuclear DNA in determining evolutionary relationships.

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- (b) Giant pandas, *Ailuropoda melanoleuca* and red pandas, *Ailurus fulgens* are both mammals which are native to China. It was long believed that giant pandas and red pandas were close evolutionary relatives and there is much evidence to support this hypothesis.

- Both animals live in similar habitats.
- Both have digestive systems similarly adapted to their bamboo diet.
- Both have a sixth digit, known as a pseudo-thumb, which they use to grip and shred bamboo shoots.



Fig. 3.2

With the advent of DNA sequencing techniques, it has been possible to compare the DNA of different species to confirm how closely related they are to each other. The table below shows mtDNA sequences from four species of mammals including that of the giant panda and the red panda.

Table 3.1

species	mtDNA codes
black bear	ATTGGAGCAGACTTA.....
giant panda	ATTGGCACTAATCTA.....
red panda	ATTGGAACTAATCTT.....
raccoon	ATCGGACTAATCTT.....

- (i) Use Table 3.1 to identify which of the species is most closely related to the red panda.

Explain your answer.

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.....[2]

In the study, mtDNA base pairs were analysed.

The following number of differences were found between the DNA of the giant panda and that of the other species analysed:

red panda 17
black bear 12
raccoon 21

An estimate of the mutation rate for the mtDNA sequence analysed is 3.95×10^{-7} mutations per year.

- (ii) Calculate how many years ago the giant panda and the red panda last shared a common ancestor. Give your answer in standard form, to two significant figures.

..... [2]

- (c) In addition to mtDNA sequences, information based on the presence of shared characteristics in various organisms can also be used to show relatedness and to draw phylogenetic trees.

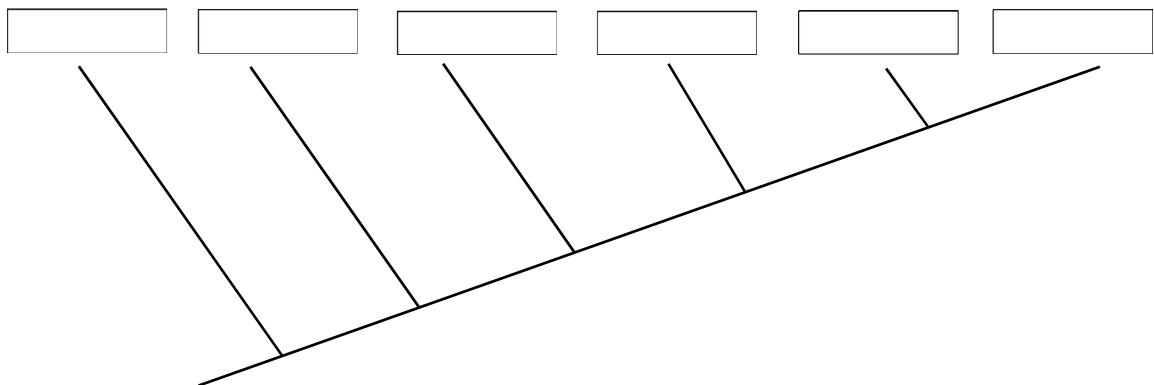
Table 3.2 shows certain characteristics present (+) or absent (-) in 6 organisms.

Table 3.2

	segmented	jaws	hair	placenta	multicellular	limbs
amoeba	-	-	-	-	-	-
cat	+	+	+	+	+	+
earthworm	+	-	-	-	+	-
kangaroo	+	+	+	-	+	+
lizard	+	+	-	-	+	+
sponge	-	-	-	-	+	-

Using the data, label the cladogram with the names of the organisms.

[1]



[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a) Apart from enzymes, describe the significance of various types of proteins in the regulation of processes and responses in cell. [15]

- (b) Compare gene mutation with chromosomal aberration. [10]

[Total: 25]

- 5 (a) Discuss how pathogens (apart from the Severe Acute Respiratory Syndrome Coronavirus 2), lifestyle and genetics can cause diseases in humans. [15]

- (b) Describe and explain the difference between continuous and discontinuous genetic variation. [10]

[Total: 25]

End of Paper